

Objectives

- 1) Compensated vs. decompensated cirrhosis
- 2) Significance of MELD-Na score
- 3) Evaluation of acute decompensation
- 4) Management of common complications of cirrhosis

“The cirrhosis one-liner”

1. Cause of cirrhosis

EtOH (60-70%)
NASH (10%)
HCV/HBV (10%)
Hemochromatosis
Uncommon (AIH, PSC, PBC)

2. Compensated?

yes

no complications
med survival: >12 y

no

+ complications
med survival: < 6 mos

3. Baseline MELD-Na score?

Cr, Tbili, INR, Na	MELD-Na	3 month mortality
	20	3-4%
	30	27-32%
	40	65-66%

4. History of complications?

ACUTE DECOMPENSATION			
↑ MELD-Na or development of below complications			
COMPLICATION	CAUSE/ PRECIPITANT	EVALUATION	MANAGEMENT
1			
2			
3			
4			
5			

Guideline Update (2021–2023)

- MELD 3.0 replaced MELD-Na for US liver allocation (OPTN, 2023). It adds serum albumin and female sex and re-weights Na/bilirubin/INR/creatinine — modestly lowers waitlist mortality and improves access for women. Where the board says "MELD-Na," current allocation uses MELD 3.0.
- Hepatorenal syndrome: terlipressin is FDA-approved (Sept 2022) and AASLD-recommended FIRST-LINE for HRS-AKI (with albumin); octreotide/midodrine + albumin remains the alternative where terlipressin is unavailable. Type 1 HRS is now termed HRS-AKI (ICA 2015 / AASLD 2021).
- Baveno VII (2022): in COMPENSATED cirrhosis with clinically significant portal hypertension (CSPH), carvedilol (the NSBB of choice) reduces first decompensation — chiefly ascites — and mortality (PREDESCI). A prevention lever upstream of this board's decompensation focus.
- Nomenclature (2023): NASH → MASH and NAFLD → MASLD; "MetALD" for metabolic dysfunction + moderate alcohol. Read the board's "NASH (10%)" cause as MASH.
- Unchanged & current: diagnostic paracentesis for everyone admitted with cirrhosis + ascites; SAAG; SBP = cefotaxime/ceftriaxone + albumin; lactulose/rifaximin for HE; ceftriaxone + octreotide + PPI + EGD for variceal bleed.

Take-Home Points

- Decompensation = a rise in MELD (now MELD 3.0) or a new complication (ascites, variceal bleed, HE, SBP, HRS). Median survival falls sharply from compensated (~years) to decompensated disease.
- Work up acute decompensation for overlapping precipitants: infection/SBP, GI bleeding, and portal vein thrombosis — diagnostic paracentesis, look for bleeding, and RUQ US with Doppler.
- Essentially EVERYONE admitted with cirrhosis and ascites gets a diagnostic paracentesis (SAAG, cell count, culture). SBP = ANC $\geq 250 \rightarrow$ cefotaxime/ceftriaxone PLUS albumin.
- HRS-AKI: hold diuretics/nephrotoxins, give albumin, and treat with terlipressin (first-line) or octreotide/midodrine + albumin. HE: lactulose + rifaximin. Variceal bleed: ceftriaxone, octreotide, PPI, urgent EGD.
- Risk-stratify and refer for transplant using MELD 3.0; in compensated cirrhosis with CSPH, carvedilol can prevent the first decompensation.

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Adapted from the TeachIM original "Management of Decompensated Cirrhosis" (published Jan 2020).

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- Optimization (June 2026): surgical correction — cause-of-cirrhosis list "Hematomachrosis" → "Hemochromatosis". All original click-to-reveal animations and graphics on the interactive board preserved.
- Added: 2021–2023 guideline-update slide (MELD 3.0, terlipressin for HRS-AKI, Baveno VII carvedilol/CSPH, MASLD/MASH) + take-home slide. Companion: Decompensated Cirrhosis learner handout (Management-of-Decompensated-Cirrhosis-learner-handout.pdf) — reviewed, no changes needed.

Medical content should be clinician-reviewed before teaching.